

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	35.3	16.8	2.4	0.6	0.2	44.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	35.4	15.6	2	0	0.1	46.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	36.1	15.8	2.4	0.5	0.2	45
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	34.5	16	1.7	0	0.2	47.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	35.8	16.2	2.5	0.6	0.2	44.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	36.2	14.8	1.9	0.1	0.1	46.9
U2 - Efficacy additive alpha prior Beta(1, 1)	35.2	16.1	2.5	0.5	0.2	45.5
U3 - Efficacy additive alpha prior Beta(1, 1)	36.7	14.4	1.9	0	0.2	46.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.6	6.3	2.5	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.1	5.9	2.3	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.6	6.3	2.5	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.2	5.9	2.3	1.3	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.7	6.3	2.5	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.2	5.8	2.3	1.3	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	14.6	6.3	2.5	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.2	5.9	2.3	1.3	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.4	0.8	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.3	0.7	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.5	0.7	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.3	0.7	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.5	0.8	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.2	0.7	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.5	0.7	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	3.3	0.7	0.6	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.2	18.3	39.3	13.9	2.3	1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.8	19.1	35.4	11.9	1.8	1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.8	18.5	37	13.9	2.8	1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.2	18.7	34.7	11.7	1.8	0.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.2	17.2	38.1	12.3	3.1	1.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	33.8	17.1	35.3	10.8	2.1	0.9
U2 - Efficacy additive alpha prior Beta(1, 1)	29.1	16	38.1	12.6	3.1	1.1
U3 - Efficacy additive alpha prior Beta(1, 1)	33.6	17.8	34.9	10.9	1.8	1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	7	9	3.8	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.8	7.2	8.7	3.5	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.3	6.9	8.8	3.8	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10	7	8.5	3.5	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	6.7	8.8	3.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.1	6.8	8.7	3.4	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.4	6.7	8.7	3.9	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.2	6.8	8.6	3.5	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.2	1.7	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.2	1.6	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.2	1.7	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.2	1.5	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.2	1.6	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.2	1.5	0.8	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.2	1.6	1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.2	1.5	0.8	0.2	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	46.7	26.9	12.3	2.1	2.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.4	53.6	21.1	8.7	1.7	2.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.7	47.2	25.9	11.8	2.1	2.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.6	54.2	21	8.1	1.8	2.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.1	46	25.8	12	2.7	2.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.9	51.2	22.7	8.1	1.6	2.5
U2 - Efficacy additive alpha prior Beta(1, 1)	10.7	46.8	25.2	12.4	2.5	2.4
U3 - Efficacy additive alpha prior Beta(1, 1)	13.4	51.7	22.2	8.6	1.6	2.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.6	11.3	6.8	3.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.8	12.2	6.4	3.4	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.7	11.2	6.8	3.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.9	12	6.4	3.4	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.7	10.8	6.7	4.1	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7	11.6	6.5	3.5	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.8	10.8	6.6	4.2	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	7	11.5	6.5	3.6	1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.8	1.8	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.8	1.6	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.8	1.7	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.8	1.6	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.8	1.7	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.8	1.6	1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.8	1.7	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.8	1.6	1	0.2	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.7	6.6	17.5	52	17	0.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	7.8	16.9	52.1	13.4	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.6	5.5	19.2	52.6	15.9	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	6.7	17	50.7	14.5	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.6	7.3	18.3	50.2	17.4	0.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.1	7.4	18.2	49.4	14.6	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	7.4	6	19.5	49.8	17.1	0.2
U3 - Efficacy additive alpha prior Beta(1, 1)	10.4	7.1	17.7	49.7	14.9	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.2	4.6	6.7	10.6	3.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.5	4.6	6.8	10.7	3.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.2	4.5	6.8	10.4	4.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	4.6	6.8	10.5	3.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.2	4.5	6.8	10.1	4.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	4.6	6.9	10.2	3.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.2	4.5	6.8	10.1	4.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.6	4.6	6.9	10.1	3.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3	1.5	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3	1.5	2.4	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3	1.6	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3	1.5	2.4	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3	1.5	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3	1.6	2.4	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3	1.5	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3	1.5	2.3	0.7	

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.5	38.4	25.4	19.7	10.7	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.1	43.6	24	17.6	6.4	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.4	37.4	26.4	20.4	10.1	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.7	44.3	23.3	17.5	6.9	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.7	36.1	27.5	19.3	11.1	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.1	43.5	23.1	17.9	7.1	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	5.6	35.8	27.9	18.7	11.7	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	8.1	42.5	24.3	17.1	7.7	0.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	7.3	7	7.1	4.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	7.9	7.3	6.9	3.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.9	7.2	6.9	7.1	4.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4	7.9	7.1	6.9	4.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.9	6.9	6.9	7.1	5.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	7.7	7.1	6.8	4.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	4	6.9	6.9	7	5.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.1	7.6	7.1	6.8	4.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	3.1	1.9	2.2	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.2	2	2.2	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	3.2	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.2	1.9	2.1	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	3.1	1.8	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.3	1.8	2.2	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	3.2	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.2	1.9	2.2	1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	63.5	17.5	4.9	1.9	0.5	11.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	66.7	16.2	3.7	1.2	0.5	11.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	63.6	16.7	5	1.8	0.5	12.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	67.4	14.7	4	1.1	0.4	12.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	65.7	15.1	4.7	2.1	0.5	11.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	69.1	14.4	3.1	1.3	0.5	11.6
U2 - Efficacy additive alpha prior Beta(1, 1)	65.4	15.4	4.5	2	0.6	12.1
U3 - Efficacy additive alpha prior Beta(1, 1)	69	14.1	3.1	1.5	0.5	11.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.5	6.1	2.7	1.3	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.1	6	2.4	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.5	6	2.7	1.3	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.2	5.8	2.4	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.7	5.7	2.7	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.5	5.6	2.4	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.6	5.8	2.7	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	18.4	5.6	2.4	1.2	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.6	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.5	0.6	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.5	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.4	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.4	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.4	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.5	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.4	0.6	0.4	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.1	48.1	22.8	11.5	3.5	1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.5	52.7	20	9.1	1.7	1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.8	47.5	22	12.3	3.4	1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16	51.3	20.3	8.9	2.4	1.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.2	44.2	22.5	12.3	4.8	1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.3	48.7	19.9	9.5	2.5	1.1
U2 - Efficacy additive alpha prior Beta(1, 1)	15.5	44.4	22.6	12.1	4.4	1
U3 - Efficacy additive alpha prior Beta(1, 1)	18.8	47.9	20.8	9.3	2.2	1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.6	12.6	5.8	2.9	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8	13.5	5.3	2.5	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.7	12.3	5.8	3	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.2	12.9	5.4	2.6	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.8	11.5	6	3.3	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.3	12.3	5.6	2.7	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.8	11.6	5.9	3.3	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.4	12.4	5.5	2.7	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.4	1.3	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.5	1.1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.4	1.3	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.4	1.1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.4	1.3	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.4	1.1	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.4	1.3	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.4	1.1	0.6	0.2	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	44.6	16.2	3.1	0.9	0.3	34.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	45.9	15.4	2.3	0.9	0.3	35.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	46.2	14.9	2.6	1.1	0.3	34.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	47.4	14.5	2.1	0.9	0.2	34.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	46.6	15	2.6	1	0.4	34.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	47.5	13.7	1.9	1	0.4	35.5
U2 - Efficacy additive alpha prior Beta(1, 1)	45.8	14.8	2.8	0.9	0.3	35.4
U3 - Efficacy additive alpha prior Beta(1, 1)	47	14.3	2.1	0.8	0.4	35.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.7	6.2	2.3	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16	6	2.2	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.8	6.1	2.3	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.2	5.9	2.1	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.7	6.2	2.3	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.1	5.9	2.2	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	15.8	6.1	2.3	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	16.1	5.9	2.1	1.1	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	2.9	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	2.9	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	2.9	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	2.9	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	3	0.7	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	2.9	0.6	0.4	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.7	15.3	22.3	16.8	26.5	4.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.7	16.8	18.6	15.2	22.9	4.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.6	14.1	22.7	16.4	27.9	4.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23	15.2	19.5	14.1	23.7	4.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15	13.9	20.9	17.3	28.4	4.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.2	15.2	18.4	14.1	24.4	4.7
U2 - Efficacy additive alpha prior Beta(1, 1)	15	13.5	20.7	17.8	28.6	4.4
U3 - Efficacy additive alpha prior Beta(1, 1)	23.4	15.8	17.9	15.1	23.4	4.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7	6.1	5.9	5.5	5.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.6	6	5.8	5.4	4.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7	5.9	5.9	5.5	5.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.8	5.8	5.8	5.4	4.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.1	5.9	5.9	5.3	5.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.8	5.9	5.8	5.1	5	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.1	5.9	5.9	5.3	5.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.9	5.9	5.8	5.1	4.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.6	1.6	1.9	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.5	1.5	1.8	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.5	1.6	1.9	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.5	1.5	1.8	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.5	1.6	1.9	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.6	1.5	1.8	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.6	1.6	1.8	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.5	1.5	1.8	0.8	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.8	43.5	17.2	6.3	1.5	20.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.1	46	15.5	4.7	0.9	20.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.6	42.7	17.8	6.3	1.8	20.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.9	46	15.1	4.9	1.1	21
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.1	42.3	17	7	2.2	20.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.7	44.2	15.8	5.2	1.2	20.9
U2 - Efficacy additive alpha prior Beta(1, 1)	11	42.7	17.2	6.6	2	20.5
U3 - Efficacy additive alpha prior Beta(1, 1)	12.6	45.6	15	4.7	1	21.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	10.7	4.8	2.2	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	11.1	4.6	2	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	10.5	4.8	2.3	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	10.9	4.6	2.1	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.6	10.3	4.8	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.8	10.7	4.7	2.1	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.6	10.3	4.9	2.4	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.8	10.7	4.6	2.1	0.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	4	1.1	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	4	1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4	1.1	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4	1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4	1.1	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4	1.1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	4	1.1	0.7	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	4	1	0.6	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	66	23.5	5.3	0.5	0	4.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.7	20.4	4	0.6	0	5.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	65.8	23.9	4.4	0.5	0	5.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	69.5	21.2	3.5	0.4	0	5.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	66.4	22.9	5.2	0.6	0	4.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	71.4	19.9	2.9	0.5	0	5.3
U2 - Efficacy additive alpha prior Beta(1, 1)	66.6	22.9	4.4	0.6	0	5.5
U3 - Efficacy additive alpha prior Beta(1, 1)	72.9	18.8	2.7	0.4	0	5.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.5	7.9	3.4	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.1	7.6	3.2	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.5	7.7	3.4	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.1	7.5	3.1	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.6	7.6	3.5	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.3	7.3	3.2	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	16.6	7.5	3.5	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.4	7.3	3.1	1.3	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.5	1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.4	0.9	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.4	1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.3	0.9	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.4	1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.3	0.9	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	3.4	1	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	3.3	0.9	0.5	0.1	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.7	11.2	8.4	44.2	24.1	0.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.6	14.3	6.7	41	21.2	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.3	11.7	8.8	45.4	23.5	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.5	13.7	7.1	42.3	20.1	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.3	11.4	9.1	45.2	23.6	0.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.6	12.8	7.2	42.7	20.5	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	9.6	12.3	9.5	44.9	23.4	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	16.5	13.2	7	42.7	20.3	0.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	5.1	5.4	9.4	5.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.9	5.1	5.3	9.6	5.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	5	5.4	9.4	5.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.9	5.1	5.3	9.5	5.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	5	5.4	9.2	5.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5	5.1	5.4	9.4	5.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.6	5	5.4	9.2	5.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	5	5	5.3	9.4	5.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.1	1.5	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.1	1.5	2.3	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.1	1.5	2.4	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.1	1.5	2.3	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.1	1.5	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.1	1.5	2.3	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.1	1.5	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.1	1.4	2.3	1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.2	44.8	17.1	6.7	1.7	17.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.9	43.9	17	5.3	1.3	17.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.7	44.2	17.1	6.3	1.8	17.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.8	43.7	15.9	5.4	1.3	17.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.9	42.5	18.2	6.3	1.6	17.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16	42.9	15.3	6.2	1.3	18.3
U2 - Efficacy additive alpha prior Beta(1, 1)	13.3	42.8	17.8	7	1.2	17.9
U3 - Efficacy additive alpha prior Beta(1, 1)	16.6	42.5	16.5	5.5	1.2	17.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	9.9	5	3	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.8	10.2	4.9	2.7	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.5	9.9	5	2.9	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.8	10	4.8	2.7	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	9.6	5.1	3	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.9	9.7	5	2.8	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.6	9.5	5.1	3	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	10	9.7	5	2.8	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.2	1.3	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.2	1.2	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.2	1.3	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.2	1.2	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4.2	1.3	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.2	1.3	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	4.2	1.3	1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.1	1.2	0.9	0.2	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.6	17.6	13.7	34.4	14.9	0.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.8	18.9	11.6	30.2	11.6	0.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.5	18	14.1	34.2	14.6	0.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.4	19.4	10.7	30.7	11.9	0.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.4	17	14.6	33.6	15.5	0.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.7	17.9	12.2	30.6	12.8	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	17.8	18.8	13.8	32.9	15.7	1
U3 - Efficacy additive alpha prior Beta(1, 1)	26.5	16.8	13	30.6	12.1	1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.2	6.1	5.6	7.7	4.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.8	6	5.4	7.7	4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.3	6.1	5.6	7.6	4.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.8	6	5.4	7.7	4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.2	6	5.7	7.6	4.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.9	6	5.4	7.5	4.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.2	6	5.6	7.6	4.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.9	5.9	5.4	7.5	4.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.5	1.6	2.2	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.5	1.6	2.1	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.5	1.6	2.2	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.5	1.6	2.1	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.5	1.6	2.2	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.6	1.5	2.1	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.5	1.6	2.2	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.6	1.5	2.1	0.9	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	47.7	30.2	13.8	6.1	0.6	1.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	61.3	26.2	8	2.5	0.4	1.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	50.1	28.5	13.1	5.9	0.7	1.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	59	26.4	9.4	3.1	0.5	1.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	49.5	29.3	12.8	6.1	0.7	1.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	59.5	26.5	8.8	3.2	0.5	1.5
U2 - Efficacy additive alpha prior Beta(1, 1)	49.8	28.4	13.3	5.9	1	1.6
U3 - Efficacy additive alpha prior Beta(1, 1)	60.5	26	8.3	3.2	0.4	1.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.1	8.2	5.7	4.2	1.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.1	7.9	5.4	4	1.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.2	8.2	5.7	4.1	1.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.2	7.9	5.4	3.9	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.3	8.1	5.7	4.1	1.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.3	7.9	5.4	3.8	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.3	8.1	5.8	4.1	1.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.3	7.9	5.4	3.8	1.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.1	1.8	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	4	1.7	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	4.1	1.8	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	4	1.7	1.6	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	4.1	1.8	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	4	1.7	1.5	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	4.1	1.8	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.2	4	1.7	1.5	0.4	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	41.3	25	13.3	13.7	6.7	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	46.9	24.5	13.3	11.2	4.1	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	41.1	23.6	14.5	13.3	7.5	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	46.4	25.4	12.9	10.9	4.4	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	39.7	23.9	15.4	13.5	7.5	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	44.8	26.1	13.8	10.7	4.6	0
U2 - Efficacy additive alpha prior Beta(1, 1)	38.5	24.5	16	13.2	7.8	0
U3 - Efficacy additive alpha prior Beta(1, 1)	46.2	25.9	13.1	10.4	4.4	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.5	4.9	6	8.3	6.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	5.1	6.2	8.3	5.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.4	4.7	6	8.3	6.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.8	5	6.2	8.2	5.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	4.6	5.9	8.1	7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.8	4.8	6.2	8.2	5.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.5	4.5	5.9	8.2	6.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.8	4.8	6.2	8.2	6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	2	2.9	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	2.1	2.9	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	2	3	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	2	2.9	1.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.9	3	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	2	2.9	1.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.9	3	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	2	2.9	1.3	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.8	58.9	17.3	2.8	0.1	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.4	60	13.7	1.8	0	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.2	57.2	16.8	2.6	0.1	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.2	57.3	13.7	1.7	0	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.7	54	18.5	2.7	0	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.7	56.5	13.7	2	0	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	26.3	52.8	18.2	2.5	0	0.2
U3 - Efficacy additive alpha prior Beta(1, 1)	28.6	55.6	13.9	1.8	0	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.1	14.1	5.3	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.8	14.4	4.7	1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.3	13.6	5.4	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.2	13.9	4.7	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	12.7	5.9	1.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.4	13.2	4.9	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.6	12.6	5.9	1.6	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.5	13.2	4.9	1.3	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.7	1.2	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.7	1	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.7	1.3	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.7	1	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.7	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.6	1	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.6	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.6	1	0.4	0.1	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.6	11.3	7.5	14.5	55.1	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.1	13.5	9.1	12.3	48	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.9	10.6	9.3	15.1	55.1	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17	13.3	9.9	11.5	48.3	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10	10	9.1	15.2	55.7	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.6	13.1	8.5	12.8	49	0
U2 - Efficacy additive alpha prior Beta(1, 1)	10.5	9.6	8.6	15.8	55.5	0
U3 - Efficacy additive alpha prior Beta(1, 1)	15.9	12.4	8.5	12.9	50.3	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	4	5.1	6.8	10.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.6	4.1	5.2	6.9	10.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	4	5.1	6.7	10.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.7	4.1	5.1	6.8	10.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.9	5	6.5	11	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.7	4.1	5.2	6.6	10.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.9	5	6.6	11	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.7	4.1	5.1	6.6	10.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.5	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.5	2.5	1.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.5	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.5	2.5	1.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.4	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.5	2.5	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.5	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.5	2.5	1.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.5	59	20.7	3.1	0.1	2.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.7	61.1	16	2.5	0.1	2.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.4	59.1	21.1	2.7	0.1	2.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.4	62.2	15.4	2.5	0.1	2.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.8	58.9	20.4	2.5	0	2.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.3	58.2	16.6	2.1	0	2.8
U2 - Efficacy additive alpha prior Beta(1, 1)	17.4	58.1	19.2	2.8	0	2.5
U3 - Efficacy additive alpha prior Beta(1, 1)	20.6	58.1	16.5	2.1	0	2.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8	12.6	6.6	2.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.4	12.8	6.2	2.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8	12.6	6.6	2.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.5	12.6	6.2	2.2	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.2	12.3	6.5	2.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.6	12.5	6.1	2.2	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.2	12.2	6.6	2.4	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.6	12.5	6.2	2.2	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.6	1.7	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.6	1.6	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.6	1.7	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.6	1.6	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.6	1.6	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.6	1.6	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.6	1.7	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.7	1.6	0.8	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.8	48.5	27.5	12.1	1.8	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14	50.4	23.6	10.4	1.3	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.5	48.1	27.1	13.2	1.8	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.3	51.7	21.9	10.4	1.4	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.3	45.5	27.4	13.4	2.1	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.9	50.1	21.6	10.9	1.2	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	12.1	45.4	27.4	12.8	2	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	16.2	50	21.8	10.3	1.4	0.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.8	10	8.2	5.2	1.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.3	10.4	7.8	4.8	1.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.9	9.8	8.1	5.2	1.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.4	10.3	7.8	4.8	1.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	9.6	8	5.3	2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.5	10.2	7.7	4.9	1.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.1	9.5	8.1	5.2	2	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.6	10	7.7	4.9	1.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.9	2.3	1.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.8	2.2	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.9	2.2	1.7	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.9	2.2	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.8	2.2	1.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.9	2.1	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.8	2.2	1.8	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.9	2.1	1.6	0.5	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	57.1	1.8	0	0	0	41.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	57.9	1.5	0	0	0	40.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	58	1.2	0	0	0	40.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	57.8	1.4	0	0	0	40.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	57.8	1.2	0	0	0	41
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	58.6	0.9	0	0	0	40.5
U2 - Efficacy additive alpha prior Beta(1, 1)	58	0.9	0	0	0	41.1
U3 - Efficacy additive alpha prior Beta(1, 1)	58.7	1.2	0	0	0	40.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.2	2.5	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.4	2.5	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.3	2.4	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.4	2.4	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.3	2.4	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.5	2.3	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	20.2	2.4	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	20.5	2.3	0.2	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.1	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.1	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1.1	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1.1	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	1.1	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	1	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	1.1	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	1	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.4	62.5	14.1	3.7	0	1.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	23	60.5	11.4	3.5	0	1.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.7	61.8	14.3	3.6	0	1.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.3	60.4	11.5	3.5	0	1.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.6	59.3	14.9	3.7	0	1.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.5	57.3	13.2	3.5	0	1.5
U2 - Efficacy additive alpha prior Beta(1, 1)	20.6	59.9	14.5	3.5	0	1.5
U3 - Efficacy additive alpha prior Beta(1, 1)	23.9	58.7	12.3	3.7	0	1.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.5	13.3	5.7	2.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	13.1	5.4	2.1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.7	13.1	5.7	2.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	12.9	5.4	2.1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.7	12.9	5.8	2.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	12.8	5.5	2.1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.8	12.8	5.8	2.3	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.4	12.8	5.4	2.1	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.8	1.7	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	4.7	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.8	1.7	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	4.7	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.7	1.7	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	4.7	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.8	1.7	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	4.7	1.6	0.7	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21	4.2	0	0	0	74.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.9	3.9	0.1	0	0	75.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.3	4	0	0	0	74.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.4	3.8	0	0	0	74.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.2	4.5	0	0	0	74.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.6	4.5	0.1	0	0	73.8
U2 - Efficacy additive alpha prior Beta(1, 1)	20.6	4.9	0	0	0	74.5
U3 - Efficacy additive alpha prior Beta(1, 1)	20.6	4.8	0	0	0	74.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.9	4.8	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.9	4.7	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.9	4.9	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16	4.6	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.9	4.8	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16	4.7	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	15.9	4.8	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.9	4.6	0.7	0.1	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.8	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.7	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.8	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.7	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.8	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.7	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.8	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.7	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	2.3	1.4	3.1	56.3	36.9	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	3	4.8	58.6	29.7	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.9	1.1	3.4	56.2	37.4	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.4	2.9	4.8	58.6	30.3	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.5	1.3	4.6	55.5	37.1	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	2.8	2.6	6	57.8	30.8	0
U2 - Efficacy additive alpha prior Beta(1, 1)	1.5	1.4	4.7	55.2	37.2	0
U3 - Efficacy additive alpha prior Beta(1, 1)	3	2.4	5.8	58.4	30.4	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.1	4	12.5	7.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.2	4.1	13.1	6.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.1	4	12.4	7.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.2	4.1	12.9	6.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.1	3.9	12.1	7.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.2	4.1	12.7	6.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.1	3.9	12.1	7.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.2	4.1	12.7	7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.4	1.1	2.5	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.5	1.1	2.5	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.4	1.1	2.5	1.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.5	1.1	2.5	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.4	1.1	2.5	1.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.5	1.1	2.5	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0	2.4	1.1	2.4	1.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0	2.5	1.1	2.5	1.2	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.1	28.6	37	6.1	0.1	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.8	30.6	32	4.5	0	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.6	29.2	36.3	5.7	0.1	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	35.2	28.3	30.9	5.3	0.2	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.7	27.3	35.6	6.1	0.1	0.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	36.5	27	31	5.3	0.1	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	29.7	28.2	36	5.9	0.1	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	36.1	27.5	31	5.2	0.1	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	9	8.8	2.8	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	9.2	8.3	2.5	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.9	8.9	8.8	2.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	9	8.4	2.6	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.1	8.6	8.8	2.9	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.8	8.7	8.4	2.7	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.1	8.7	8.8	2.9	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.9	8.8	8.3	2.6	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.7	1.9	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.6	1.8	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.7	1.8	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.6	1.8	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.6	1.8	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.6	1.7	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.7	1.8	1	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.6	1.7	0.9	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.8	32.2	34.1	7.4	0.1	3.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.6	31.3	30.2	6.5	0.1	3.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	24	31.3	34.3	6.9	0.1	3.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	31.9	29.8	29.1	5.9	0.1	3.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.5	30.5	33.9	6.6	0.1	3.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.4	29.9	30.3	5.2	0.1	3.1
U2 - Efficacy additive alpha prior Beta(1, 1)	26.9	29.2	34.5	5.9	0.2	3.3
U3 - Efficacy additive alpha prior Beta(1, 1)	32.7	29.2	29	5.7	0.1	3.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	8.9	8.4	3.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.4	8.8	8.1	2.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9	8.7	8.4	3.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	8.7	8	2.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9	8.6	8.5	3.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.7	8.6	8.1	2.9	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.1	8.6	8.4	3.1	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.7	8.6	8.1	2.9	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4	1.7	1.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.9	1.6	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4	1.6	1.2	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4	1.6	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4	1.6	1.2	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4	1.6	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	4	1.7	1.2	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	4	1.6	1.1	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.6	26.6	46.7	11	0.5	0.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.7	26.4	42.6	9.1	0.4	0.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.7	25.1	46.4	11.5	0.6	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.8	26.3	42.2	8.8	0.3	0.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.6	24	47.9	10.4	0.5	0.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.3	26.8	42.3	7.8	0.2	0.6
U2 - Efficacy additive alpha prior Beta(1, 1)	18	23.8	46.8	10.1	0.6	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	22.3	25.9	43.3	7.5	0.2	0.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.6	8.2	10.2	4	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.2	8.1	10.1	3.8	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.6	8.1	10.2	4.1	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.3	8.1	10.1	3.8	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.8	7.8	10.2	4.1	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.4	8.1	10	3.7	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.9	7.8	10.2	4.1	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.5	8	10	3.6	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.7	2.1	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.7	2	1.4	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.7	2.1	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.7	2	1.4	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.7	2	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.8	1.9	1.3	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.7	2	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.8	1.9	1.3	0.2	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.7	52	9.4	0.5	0	19.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.6	51.6	7.8	0.3	0	19.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.2	54.8	8.1	0.6	0	19.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.1	51.1	7.6	0.3	0	19.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.8	53	7.7	0.5	0	19
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.3	51.5	7.3	0.2	0	19.7
U2 - Efficacy additive alpha prior Beta(1, 1)	18.4	53.5	8.6	0.5	0	19
U3 - Efficacy additive alpha prior Beta(1, 1)	21.2	52	7	0.4	0	19.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.7	12.1	4.1	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11	12	3.9	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	12	4.1	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.2	11.9	3.9	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.9	11.8	4.1	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.3	11.6	3.9	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.9	11.8	4.2	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.4	11.7	3.9	1.2	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.6	1.2	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.5	1.2	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.6	1.2	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.5	1.2	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.5	1.2	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.5	1.2	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.6	1.2	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.5	1.2	0.4	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	66.1	27.1	4.2	0.2	0	2.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.6	25.2	2.8	0.1	0	2.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	67.6	26.3	3.5	0.2	0	2.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	70.6	24	2.8	0.3	0	2.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	68.1	25.7	3.6	0.3	0	2.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	71.7	22.8	3.1	0.2	0	2.2
U2 - Efficacy additive alpha prior Beta(1, 1)	68.5	25.2	3.7	0.2	0	2.4
U3 - Efficacy additive alpha prior Beta(1, 1)	72.5	22	3.1	0.2	0	2.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.7	8.5	2.7	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.7	8.2	2.2	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.8	8.4	2.7	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.9	7.9	2.2	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18	8.1	2.8	0.7	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19	7.7	2.3	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	18	8	2.8	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.1	7.7	2.3	0.5	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3	0.9	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.9	0.7	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.9	0.9	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.8	0.7	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.9	0.9	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.8	0.7	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.8	0.9	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.8	0.7	0.2	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.2	11.9	11.2	45.3	14.4	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.9	14	9.8	38.3	10	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.1	11.5	12.5	44.3	14.6	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.4	14.4	10.2	39.7	10.3	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.5	10.6	13.4	44.7	12.8	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.7	13.8	11	39.4	10.1	0
U2 - Efficacy additive alpha prior Beta(1, 1)	18.1	10.9	13	44.4	13.6	0
U3 - Efficacy additive alpha prior Beta(1, 1)	25.5	13.5	11.7	39.5	9.8	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.3	4.8	6	10.3	4.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	4.9	5.9	10.3	4.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.4	4.7	5.9	10.3	4.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	4.9	5.9	10.3	4.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	4.6	5.9	10.2	4.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.8	4.8	5.9	10.1	4.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.4	4.6	6	10.1	4.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.8	4.8	5.9	10.2	4.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.7	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.8	1.7	2.4	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.7	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	1.6	2.3	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.6	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	1.6	2.3	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.6	2.4	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	1.6	2.3	1.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.9	35.7	26.7	26.2	6.5	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.7	42.5	25.4	20.2	4.2	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.4	33.9	26.1	26.9	6.7	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.7	40.6	26.6	19.8	4.3	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.6	32.8	27.4	26	7.2	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.1	38.7	27.3	19.7	4.2	0
U2 - Efficacy additive alpha prior Beta(1, 1)	7	32.8	27.2	26	7	0
U3 - Efficacy additive alpha prior Beta(1, 1)	9.9	38.9	27	20.1	4.1	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	11.1	7.2	5.8	2.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	12	7.1	5.3	1.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.9	10.6	7.1	6.2	2.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	11.6	7.1	5.5	1.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	9.9	7.3	6.2	2.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	10.8	7.3	5.7	2.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	4	9.8	7.2	6.3	2.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.1	10.8	7.2	5.7	2.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.6	1.5	1.3	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.4	1.1	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.6	1.5	1.3	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.4	1.1	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.4	1.3	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.4	1.2	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.4	1.3	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.4	1.2	0.6	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.6	15.1	33.3	26.2	11.8	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20	16.3	33.3	23.4	7	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.9	13.9	36.3	24.4	11.5	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.8	16.9	33.8	22.2	7.3	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.1	14.4	35.4	24.1	12	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.5	16.2	33.8	22.6	7.9	0
U2 - Efficacy additive alpha prior Beta(1, 1)	14.3	14.2	36.8	21.9	12.8	0
U3 - Efficacy additive alpha prior Beta(1, 1)	18.8	17.2	35	21	8	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	4.3	8.5	8.5	5.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.6	4.5	8.9	8.4	4.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	4.1	8.5	8.4	5.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.7	4.4	8.6	8.4	4.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	4.1	8.3	8.4	5.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	4.3	8.7	8.2	5.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	4	8.4	8.3	5.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.8	4.3	8.7	8.1	5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.9	2.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.9	2.7	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.8	2.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.8	2.7	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.8	2.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.8	2.7	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.8	2.8	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.8	2.7	1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.3	5.1	10.4	45.1	31.1	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.6	7.5	12.7	43	24.2	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7	4.9	11.1	44.6	32.4	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.3	7.6	11.9	43.7	24.5	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.1	5.5	10.7	44.2	32.5	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.7	7.8	12.3	43	25.2	0
U2 - Efficacy additive alpha prior Beta(1, 1)	7.1	5.4	11.5	43.8	32.2	0
U3 - Efficacy additive alpha prior Beta(1, 1)	11.9	7.2	12.7	41.7	26.5	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.3	3.6	5.9	10.3	6.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.7	6	10.5	6.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.4	3.6	5.9	10.1	7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.8	6	10.4	6.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.4	3.6	5.8	9.9	7.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.8	6	10.2	6.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.4	3.6	5.7	9.9	7.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.6	3.7	6	10.1	6.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.6	1.5	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.7	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.6	1.4	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.7	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.6	1.5	2.7	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.7	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.5	2.7	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.5	2.7	1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.4	12.8	44.8	22.3	5.5	0.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.9	15.2	43	20.4	3.3	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.7	13.7	46.6	21.7	5	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.8	15.6	41.8	20.6	2.9	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.7	13.8	46.6	21.3	5.4	0.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.8	16.1	42.2	19.9	2.8	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	12.2	13.7	47.8	20	6	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	19.7	14.5	44	18.5	3.1	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.1	6.3	9.6	6.2	2.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.8	6.3	9.5	6	2.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	6.1	9.6	6.2	2.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.9	6.3	9.4	6	2.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	6.1	9.5	6.1	3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.9	6.2	9.4	5.9	2.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.3	6.2	9.5	6	3	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.9	6.2	9.5	5.8	2.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.4	2.1	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.5	2	1.9	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.4	2.1	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.4	2	1.9	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.4	2	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.4	2	1.9	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.4	2	2	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.4	2	1.9	0.6	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.2	40.3	31.5	14.3	3.7	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.8	46.6	27.7	9.4	2.5	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.6	38.3	31.2	15.2	3.7	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.9	44.8	27.9	9.8	2.6	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.6	35.9	31.1	16	4.4	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.9	42.7	28.3	10.5	2.6	0
U2 - Efficacy additive alpha prior Beta(1, 1)	12.5	35.9	31.3	16.2	4.1	0
U3 - Efficacy additive alpha prior Beta(1, 1)	16.7	42	27.8	10.8	2.7	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	11.2	8.5	4.4	1.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	12.2	7.9	4	1.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	10.7	8.4	4.6	1.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.8	11.7	8.1	4	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	10	8.5	4.9	2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.9	11	8.3	4.2	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.6	10	8.4	5	2	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.9	11	8.3	4.2	1.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3	1.9	1.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3	1.8	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3	1.9	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.1	1.8	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.1	1.8	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.1	1.8	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3	1.8	1.3	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.1	1.8	1.1	0.3	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.8	48.7	15.4	7.8	1.6	5.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25	47.6	13	7.2	1.2	6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.9	46	16.2	8.1	1.8	6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.1	47.8	13.6	7.2	1	6.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.8	45.4	17.2	8.1	2	6.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.3	46.8	13.3	7.4	1.4	6.8
U2 - Efficacy additive alpha prior Beta(1, 1)	20.7	46.4	16.4	7.9	2.1	6.5
U3 - Efficacy additive alpha prior Beta(1, 1)	24.3	47.4	13	7.4	1.5	6.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.7	10.5	5.5	3.5	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	10.3	5.4	3.4	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	10.4	5.6	3.5	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.3	10.2	5.4	3.4	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	10.1	5.6	3.6	1.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	10.2	5.4	3.4	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.8	10.1	5.7	3.5	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.5	10.1	5.4	3.4	1.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.6	1.6	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.6	1.5	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.7	1.5	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.6	1.5	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.7	1.5	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.7	1.5	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.7	1.6	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.7	1.5	1.1	0.3	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.9	57.1	16.3	3.8	0.5	3.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.7	55.3	13	3.1	0.5	3.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.5	54.3	16.2	4.1	0.6	3.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.3	54	13.5	3.1	0.7	3.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.1	52.7	16.3	4.7	0.8	3.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.4	54.2	12.9	2.7	0.5	3.3
U2 - Efficacy additive alpha prior Beta(1, 1)	22.4	53	16.1	4.3	0.7	3.5
U3 - Efficacy additive alpha prior Beta(1, 1)	25.6	54	13.3	3	0.7	3.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.4	11.7	5.7	3	0.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.1	11.4	5.4	3	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.5	11.6	5.8	3.1	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	11.3	5.5	2.9	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.6	11.3	5.8	3.1	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	11.3	5.5	2.8	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.7	11.3	5.8	3.1	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.4	11.2	5.5	2.8	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	5	1.7	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.9	1.6	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	5	1.7	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	5	1.6	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	5.1	1.7	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	5	1.6	1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	5.1	1.7	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	5	1.6	1	0.2	